

LinkedIn

www.linkedin.com/in/mohamed-amara-1b45752b4

Portfolio (to see other projects)

<https://mohamedamara.netlify.app>

PROJECTS

Human-Powered Washing Machine

Mar 2025

- EDesigned and prototyped a **porable, electricity-free laundry system** that uses a pedal-powered drivetrain to clean 5kg of clothing, aimed at supporting off-grid and sustainable living
- Developed 3D models using **SolidWorks**, incorporating ergonomic and anthropometric data to optimize the drum, frame, and seat for 50th percentile users
- Performed **load, torque, and fatigue** calculations on shafts and transmission components, achieving a safety factor of 2
- Engineered a dual-ratio chain and sprocket system with a gear shifter to enable gear changes between washing and spin-dry cycles, ensuring optimal torque-speed balance
- eSelected and validated materials such as aluminum 6061-T6, SAE 1144 steel, and 6300-series bearings based on strength-to-weight ratio, durability, and corrosion resistance

EXPERIENCE

Software Engineer Intern

Jan 2025 – Apr 2025

CoinWa – Waterloo, ON

- Developed a cross-platform financial literacy app using **Flutter** and **Dart** to teach teenagers budgeting and saving skills through gamified modules
- Built backend services with **Node.js** and **MongoDB** to manage authentication, user data, and progress
- Designed responsive **UI** layouts for various screen sizes using Flutter; used **Canva** for wireframes and visual assets
- Implemented login, email verification, and password recovery features ensuring **secure** user access
- Debugged and **optimized** app performance using Flutter DevTools, improving load times and eliminating rendering issues

Software Developer Intern

Feb 2023 – Jun 2023

Western University Outreach – London, ON

- Collaborated with the Engineering Department Chair to **develop** a teacher-course assignment **program**
- Developed and implemented a Java-based tool to **optimize** teacher-course allocation, improving **efficiency** by **80%** by **automating** assignments based on teacher qualifications and availability
- Provided technical **support** and training to staff, ensuring seamless program usage
- Documented findings and wrote **reports** for clients and regulators, ensuring compliance with **standards**

SKILLS AND ABILITIES

- **Programming Languages:** MATLAB, C++, C, JavaScript, Node, Java, Python, HTML
- **Robotics & Embedded Systems:** ROS2, Arduino, PLC, FPGA, Sensors and Actuators
- **CAD and Simulation Tools:** AutoCAD, SolidWorks, MySQL, GIS, GitHub, GitLab
- **Soft Skills:** Problem-solving, Team Collaboration, Time Management, Adaptability

EDUCATION

University of Waterloo – Waterloo, ON

Present

Candidate for BAsC in Mechatronics Engineering Department

- **Relevant Courses:** Sensors and Instrumentations, Engineering Design, Digital Logic, Solid Mechanics, Circuits, Computer Structures and Real-Time Systems